NLP project: Frameworks for Chinese Spoken Language Understanding

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1. Abstract

Spoken language understanding (SLU) is an emerging field in between the areas of speech processing and natural language processing. It is an critical part of conversation system, which aims to extract users' intention and s the foundation for follow-ups.In this work, we mainly focus on how to improve the effect of information extraction, we not only sophisticatedly train the given baseline model, but also try many recently widely-concerned models in SLU, such as Bi-LSTM-CRF, Bert-CRF and Transformer-CRF. The accuracy of the baseline model on given dataset reaches 71.06%, and the accuracy of our best model Transformer-CRF reaches 87.49%.

2. Problem Definition

Fig. 1(a) shows a diagram of Spoken dialogue system, which includes five parts. Our task focuses on the SLU part, which takes words as input and outputs their semantics. Specifically, the task can be seen as a multi-class classification, which assigns BIO labels to input. The BIO annotation labels each word as B-X, I-X, or O, where X is the category of a fragment. In general, B-X means this word is at the beginning of a fragment, and the fragment belongs to category X,I-X means this word is at the end of a fragment and the fragment belongs to category X, and O means this word doesn't belong to any valid fragment, which indicates no semantic information.

3. Method

3.1. Base Processing

In problem definition section, we introduce the BIO annotation:B-X,I-X and O, where X means the type of fragment.In our dataset, there are 2 types of action and 18 types of slots. We add a type of [PAD] to represent the padding word, so there are totally 74 (36×2+1+1) tags in our dataset.We then convert the SLU problem into a multi-class classification task.

In our model, all of the words in the input sequences are denoted by indexes and embedded as vectors by pretrained

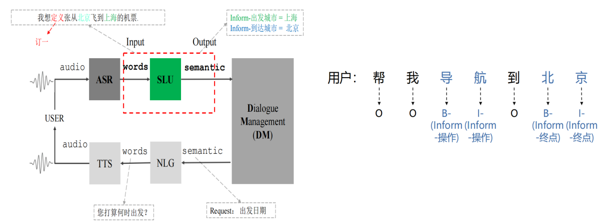

Figure 1. (a) A diagram of Spoken dialogue system. (b) An example of BIO annotation. In (b), Act-slot can be combined as a tag and then we can use the B-[Act plot] and I-[Act plot] tags to determine the value.

word embedding. Since the length of input sequence in a batch should be uniform, we pad the sequence to the max length of the input sequence in a batch and construct a mask to distinguish the original sequence and the padding part.

3.2. Models

3.2.1 Bi-RNN

In baseline model, bidirecrional RNN acts as the encoder. For each word in the input sequence, each layer computes the following function: where h_t is the hidden state at time t, x_t is the input at time t, and h_t-1 is the hidden state of the previous layer at time t 1 or the initial hidden state at time o.

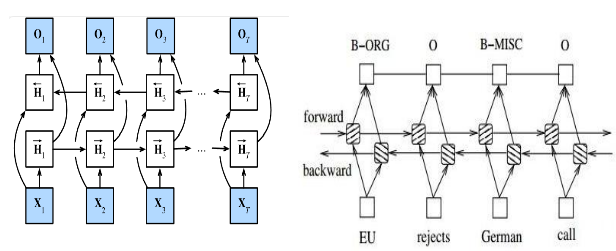

Figure 2. (a)Architecture of Bi-RNN. (b)Architecture of Bi-LSTM

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3.2.2 Bi-LSTM

In baseline model, we replace bi-RNN with bidirectional long short term memory (LSTM) as the encoder. For each word in the input sequence, each layer computes the following function:

$$
\begin{aligned}
i_t &= sigmoid(W_{ii}x_t + b_{ii} + W_{hi}h_{t-1} + b_{hi}) \\
f_t &= sigmoid(W_{if}x_t + b_{if} + W_{hf}h_{t-1} + b_{hf}) \\
o_t &= sigmoid(W_{io}x_t + b_{io} + W_{ho}h_{t-1} + b_{ho}) \\
g_t &= tanh(W_{ig}x_t + b_{ig} + W_{hg}h_{t-1} + b_{hg}) \\
c_t &= f_t c_{t-1} + i_t g_t \\
h_t &= o_t * tanh(c_t)
\end{aligned}
\tag{1}
$$

where h_t is the hidden state at time t, c_t is the cell state at time t, x_t is the input at time t, h_t1 is the hidden state of the layer at time t 1 or the initial hidden state at time o. i_t, f_t, g_t, ot are the input, forget, cell, and output gates, respectively.

3.2.3 Bi-LSTM-CRF

Conditional random field considers the correlation between labels, thus adding some constraints to prevent undesirable conditions, such as I-Y after B-X. We add the CRF layer after Bi-LSTM layer to improve model's performance.

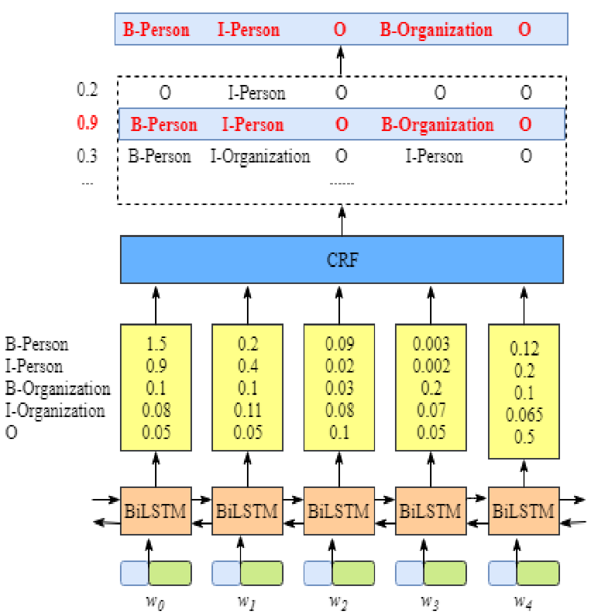

Figure 3. Architecture of Bi-LSTM-CRF

3.2.4 BERT-CRF

Bidirectional Encoder Representation from Transformers(BERT) [2] is one of the most successful models in NLP in recent years, and there are many pretrained models for BERT.

We use BERT to replace the Bi-LSTM layer in Bi-LSTM-CRF. The pretrained model we choose is chinese-bert-wwm-ext, which is pretrained on Chinese Wikipedia and other general data including Q&A conversations and news. Besides, in word to vector step, we also used the word embedding from BERT.

3.2.5 Transformer-CRF

The Transformer model [1] has emerged as another landmark in the field of NLP in recent years. In our variant of this model, the encoder component is substituted with a streamlined Transformer architecture. We feed the embedded `input_id` and `tag_id` directly into the Transformer. To preclude the possibility of future tokens

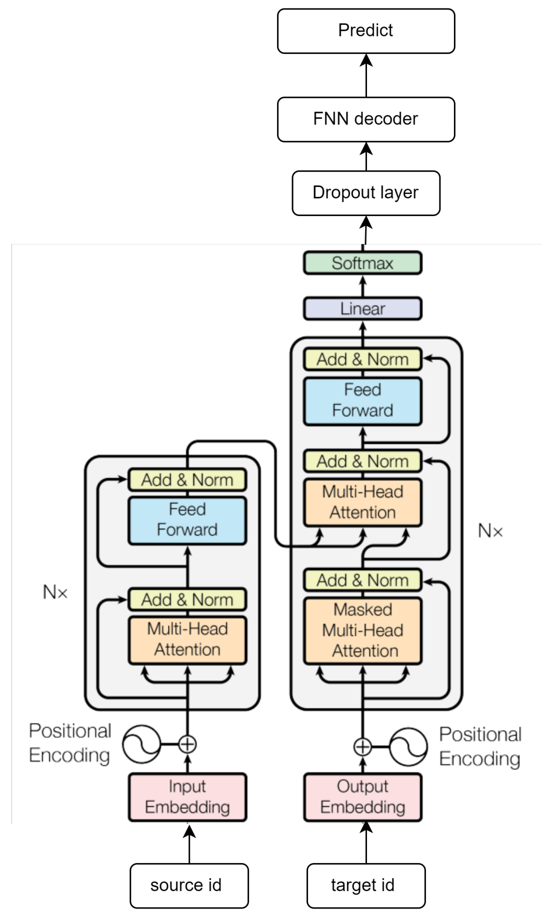

Figure 4. Transforer Stucture in SLUTagging

influencing the prediction of current ones, we employ the `transformer.generate_square_subsequent_mask` method. This ensures that, for any given position within the sequence, only the preceding elements are utilized during prediction.

The remainder of our baseline model persists without alteration. Furthermore, in the decoding phase—translating model outputs into labels—we optionally incorporate a CRF (Conditional Random Fields) layer to refine our evaluations.

3.3. Anti Noise

We define a function `anti_noise_prediction` to refine the predictions based on the decoder's output. This function's primary role is to correct the results of the SLU system to align with the predefined vocabulary specified in `ontology.json`.

It checks these predictions with a set of predefined location names (e.g. 'poi name', 'start name'. If the label in the prediction is not 'value' but one of the above positions or categories, it tries to correct it by finding the closest valid slot (slot) value.

The `LabelVocab` class retrieves the pinyin set `pinyin_set` of all POI names from the file `/data/lexicon/poi_name.txt` along with the mapping dictionary `map_dic`. It checks if the noise is directly in `pinyin_set`, and if so, returns the corresponding value directly.

If not, it calculates the Jaccard similarity between the noise and each element in `pinyin_set`, selecting the one with the highest similarity.

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}$$

This method is a heuristic search that assumes that pinyin pairs with more shared elements are more similar.

4. Experiments

4.1. Model Comparison

Model	Accuracy	Precision	Recall	F1 Score
Bi-RNN	70.83	79.91	73.93	76.75
Bi-LSTM	71.06	80.85	73.51	77.01
GRU	71.95	81.37	75.18	78.15
BERT	76.31	82.51	80.02	81.25
Transformer	**87.49**	99.88	**88.32**	**93.75**
Bi-RNN-CRF	72.18	84.35	74.77	79.27
Bi-LSTM-CRF	73.28	85.29	76.11	79.53
GRU-CRF	73.88	86,32	76.12	79.89
BERT-CRF	78.99	81.25	78.62	79.92
Transformer-CRF	**87.49**	99.88	**88.32**	**93.75**

Table 1. the evaluation results of all the models

- Tab1 displays models' performance on the given dataset. Transformer and Transformer-CRF equally perform best on the given dataset. BERT and Transformer, both with CRF layer and without CRF layer, outperform RNN and LSTM greatly. Besides, the results show that models (except Transformer) with CRF layer outperform the corresponding models without CRF layer about 1% in accuracy. Such improvements suggest that CRF layer is valid for better prediction performance for spoken language understanding tasks.

  The reason why CRF layer doesn't improve the Transformer's performance may be the given dataset is small, only Transformer can learn the constraints between labels. If we want to study whether Transformer has the ability to learn the constraints, we need larger and more complex dataset.

- The results also show that Baseline-GRU beats all the other baselines.GRU is an efficient encoder compared to LSTM RNN, so we use GRU in subsequent references to the transformer and BERT.

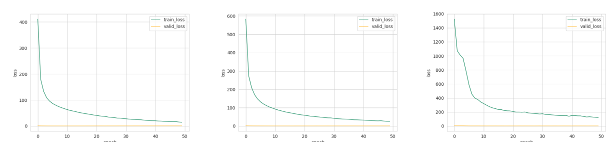

Figure 5. The training and validation loss of the Bi-RNN-CRF, Bi-LSTM-CRF and BERT-CRF.

4.2. Impact of Noise

Model	Accuracy	Precision	Recall	F1 Score
GRU	71.95	81.37	75.18	78.15
GRU anti-noise	72.13	81.02	77.23	78.32
Transformer	**87.49**	99.88	**88.32**	**93.75**
Transformer anti-noise	**87.52**	99.32	**89.12**	**94.23**

5. Conclusion

In this work, we mainly focus on how to improve the effect of information extraction, we not only sophisticatedly train the given baseline model, but also train BERT and Transformer on the given dataset. Besides, we add a CRF layer to learn the constraints between labels for better performance. The accuracy of the baseline model on given dataset reaches 71.06%, and the accuracy of our best model Transformer-CRF reaches 87.49%.

6. Outlook

6.1. Pinyin-Based Noise Mitigation

To counteract the noise stemming from misinterpretations in our anti-noise module, we employ the Jaccard Similarity Index to ascertain the value that most closely aligns in pinyin similarity. For more intricate applications, the adoption of sophisticated string or pinyin similarity algorithms might be warranted. These could include methods like the edit distance—specifically, the Levenshtein distance—or phonetic and rhyming schemes for a more nuanced matching.

6.2. Data Augmentation

To help the model generalize better to new and unseen data, especially if the training data is limited or skewed, we added data argumentation. Specifically, data argumentation methods include speech transformation (changing pitch, speed, or accent), text expansion (using synonym replacement, sentence reconstruction), noise injection (adding background noise), and data synthesis (generating entirely new speech or text data).

References

[1] Niki Parmar Ashish Vaswani, Noam Shazeer. Attention is all you need. *Neural Information Processing Systems*, 2017. 2

[2] Kenton Lee Kristina Toutanova Jacob Devlin, Ming-Wei Chang. Bert: Pre-training of deep bidirectional transformers for language understanding. 2018. 2